

# XRweld: An In-Situ Extended Reality Platform for Welding Education

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## ABSTRACT

Craft-based manufacturing trades such as metal welding require extensive hands-on training and mentorship to gain embodied knowledge, muscle memory, and experiential learning. As such, there is an opportunity to leverage spatial computing platforms to monitor welding behaviors, provide feedback, and provide assistive value through an extended reality (XR) system. Our demo leverages an XR headset and controller that have been integrated into a fully-functional welding helmet and torch, allowing users to receive in-situ, real-time feedback while actively welding. This system also allows us to explore other aspects of the embodied learning of welding, including the analysis of biometric data, performance analysis, and the inclusion of meditation to further augment the welding educational experience.

### ACM Reference Format:

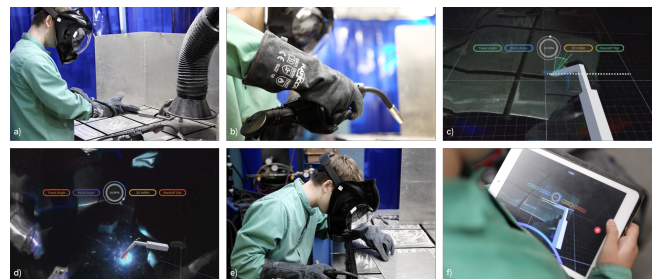
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## 1 INTRODUCTION

Despite the prevalence of welding in construction, manufacturing, and creative arts, training novice welders is a time-intensive and challenging process. To build welding expertise, learners must build deep embodied knowledge such as hand-eye coordination, muscle memory, and sound identification. Traditionally, this education is done through apprenticeship-style educational experiences that

require the direct attention and mentorship of a senior welder. We present a system that assists with this deficit, by providing realtime, in-helmet, and in-situ feedback and response to augment the educational experience. Prior works in the areas of extended reality and welding create isolated, fully immersive simulations of welding, rather than an augmentation of actual welding. By enabling users to experience the real behaviors, forces, sounds, and overall experience of an active weld, with the spatial and biometric analysis of our system, we provide a novel in-situ experience. Our prior publications on earlier versions of this work [Johnson et al. 2023], [Chen et al. 2024] include more details on related works, citations, our co-design process and hardware development, and specifics about the system setup and training.

## 2 EXPERIENCE OVERVIEW



A major feature of our work is the ability to provide XR augmentation during active welding. However, due to the safety concerns and logistical issues related to having an active welding arc in a demo-style event, our system will be configured to function with 'mock' or simulated welds. Participants will still be able to experience the torch tracking, active feedback, and biometric analysis of our system, without any risk of a live spark.

As depicted in the above figure, participants will experience our bespoke XR hardware system that integrates the welding setup. They can calibrate and plan their weld line using a point-to-point

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guideline function, and experience feedback on their position, speed, angle and behavior as they create a simulated weld. Throughout the experience, their breathing pattern is monitored to assist them with practicing mindful breathing as they accomplish their welding tasks. Realtime point-of-view feeds from the helmet can then be displayed on auxiliary monitors, allowing an instructor or audience to review or analyze the learner's performance. We intend to have multiple synchronous welding setups, highlighting our system's ability to have a scalable impact on training the manufacturing workforce and allowing many conference attendees to engage with our immersive experience.

## REFERENCES

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